

Exercise #1

16. January 2023

Exercises marked with a (J) should be handed in as a Jupyter notebook.

Optional exercises will not be corrected.

Problem 1. (Taylor Polynomials)

- a) Compute all Taylor polynomials of $f(x) = x^4 + 2x^3 + x^2 + 5$ around $x_0 = 1$
- b) Compute the Taylor series of $g(x) = \ln(1+x)$ around $x_0 = 0$

Solution.

- a) We have to compute $T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$ for all $n \in \mathbb{N}$. We first need the derivatives of f and their values at $x_0 = 1$.

We compute

$$\begin{aligned} f(1) &= 1 + 2 + 1 + 5 = 9 \\ f'(x) &= 4x^3 + 6x^2 + 2x & f'(1) &= 4 + 6 + 2 = 12 \\ f''(x) &= 12x^2 + 12x + 2 & f''(1) &= 12 + 12 + 2 = 26 \\ f'''(x) &= 24x + 12 & f'''(1) &= 36 \\ f^{(4)}(x) &= 24 = f^{(4)}(1) \end{aligned}$$

And all higher derivatives $f^{(k)}(x)$, $k > 4$ are zero.

We obtain for the Taylor polynomials

$$T_0(x) = f(1) = 9$$

$$T_1(x) = T_0(x) + \frac{f'(x_0)}{1!}(x - x_0) = 9 + 12(x - 1) = 12x - 3$$

$$T_2(x) = T_1(x) + \frac{f''(x_0)}{2!}(x - x_0)^2 = 12x - 3 + \frac{26}{2}(x - 1)^2 = 12x - 3 + 13(x^2 - 2x + 1) = 13x^2 - 14x + 10$$

$$T_3(x) = T_2(x) + \frac{f'''(0)}{3!}(x - 1)^3 = 13x^2 - 14x + 10 + \frac{36}{6}(x^3 - 3x^2 + 3x - 1) = 6x^3 - 5x^2 + 4x + 4$$

$$T_4(x) = T_3(x) + \frac{f^{(4)}(0)}{4!}(x - 1)^4 = 6x^3 - 5x^2 + 4x + 4 + (x - 1)^4 = f(x)$$

$$T_n(x) = T_4(x) = f(x) \quad \text{for } n > 4$$

b) since $x_0 = 0$ we have $x = x_0 + h = h$ and we hence have to compute

$$g(x) = \sum_{k=0}^{\infty} \frac{g^{(k)}(0)}{k!} x^k.$$

We have

$$g(0) = \ln 1 = 0,$$

$$g'(x) = \frac{1}{1+x} \qquad g'(0) = 1,$$

$$g''(x) = -\frac{1}{(1+x)^2} \qquad g''(0) = -1,$$

$$g'''(x) = \frac{2}{(1+x)^3} \qquad g'''(0) = 2,$$

$$g^{(k)}(x) = \frac{(k-1)!(-1)^{k-1}}{(1+x)^k} \qquad g^{(k)}(0) = (k-1)!(-1)^{k-1}.$$

So the Taylor series of g reads

$$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}(k-1)!}{k!} x^k = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} x^k$$

Problem 2. (Taylor expansion error)

Show that for a function $f \in C^l([0, 1]; \mathbb{R})$, we have the following expression for the remainder of the Taylor expansion (about 0, evaluated at 1),

$$f(1) - \sum_{k=0}^{m-1} \frac{f^{(k)}(0)}{k!} = \int_0^1 \frac{m}{m!} s^{m-1} f^{(m)}(s-1) ds,$$

for every $m \leq l$.

Hint. Use induction. Check that it holds for $m = 1$, and then try to manipulate the expression,

$$f(1) - \sum_{k=0}^m \frac{f^{(k)}(0)}{k!} = f(1) - \sum_{k=0}^{m-1} \frac{f^{(k)}(0)}{k!} - \frac{f^{(m)}(0)}{m!},$$

so that it is equal to $\int_0^1 \frac{m+1}{(m+1)!} s^m f^{(m+1)}(s-1) ds$.

Solution.

We follow the hint:

First we check that it holds for $m = 1$:

$$f(1) - f(0) = \int_0^1 f^{(1)}(1-s) ds = \int_0^1 f^{(1)}(y) dy = f(1) - f(0).$$

Next we show that it holds for $m+1 \leq l$, using that it holds for m . That is, we want to show,

$$f(1) - \sum_{k=0}^m \frac{f^{(k)}(0)}{k!} = \int_0^1 \frac{m+1}{(m+1)!} s^m f^{(m+1)}(s-1) ds.$$

From the hint, we decompose the left hand side such that,

$$f(1) - \sum_{k=0}^m \frac{f^{(k)}(0)}{k!} = f(1) - \sum_{k=0}^{m-1} \frac{f^{(k)}(0)}{k!} - \frac{f^{(m)}(0)}{m!} = \int_0^1 \frac{m}{m!} s^{m-1} f^{(m)}(1-s) ds - \frac{1}{m!} f^{(m)}(0).$$

Now, we use that,

$$f^{(m)}(1-s) = f^{(m)}(0) + \int_0^{1-s} f^{(m+1)}(y) dy,$$

in addition to the fact that,

$$\int_0^1 s^{m-1} = \frac{1}{m},$$

to get

$$\begin{aligned} & \int_0^1 \frac{m}{m!} s^{m-1} f^{(m)}(1-s) ds - \frac{1}{m!} f^{(m)}(0) \\ &= \int_0^1 \frac{m}{m!} s^{m-1} \left(f^{(m)}(0) + \int_0^{1-s} f^{(m+1)}(y) dy \right) ds - \frac{1}{m!} f^{(m)}(0) \\ &= \int_0^1 \frac{m}{m!} s^{m-1} \int_0^{1-s} f^{(m+1)}(y) dy ds. \end{aligned}$$

For the last double integral we can change the order of integration. We are integrating over the domain, $s \in [0, 1]$, $y \in [0, 1-s]$, which is the same as integrating over $y \in [0, 1]$, $s \in [0, 1-y]$. So we have,

$$\begin{aligned} & \int_0^1 \int_0^{1-s} \frac{m}{m!} s^{m-1} f^{(m+1)}(y) dy ds \\ &= \int_0^1 \int_0^{1-y} \frac{m}{m!} s^{m-1} f^{(m+1)}(y) ds dy \\ &= \int_0^1 \frac{m+1}{(m+1)!} (1-y)^m f^{(m+1)}(y) dy \\ &= \int_1^0 \frac{m+1}{(m+1)!} s^m f^{(m+1)}(1-s) (-ds) \quad (\text{change of variable } s = 1-y) \\ &= \int_0^1 \frac{m+1}{(m+1)!} s^m f^{(m+1)}(1-s) ds, \end{aligned}$$

which was what we wanted to show.

Problem 3. (Numerical differentiation)

We want to find a difference formula of the form

$$u''(x) \approx \frac{1}{h^2} \left(a_1 u(x) + a_2 u(x-h) + a_3 u\left(x - \frac{1}{2}h\right) \right),$$

where a_1 , a_2 and a_3 are constants to be determined.

Find the coefficients a_1 , a_2 and a_3 which makes this scheme convergent. Find an expression for the error term.

Solution.

We proceed with the expansion of the terms

$$\begin{aligned}
 u(x-h) &= u(x) - hu'(x) + \frac{h^2}{2}u''(x) - \frac{h^3}{6}u'''(x) + O(h^4) \\
 u\left(x - \frac{h}{2}\right) &= u(x) - \frac{h}{2}u'(x) + \frac{h^2}{8}u''(x) - \frac{1}{6}\left(\frac{h}{2}\right)^3u'''(x) + O(h^4)
 \end{aligned}$$

Plugging into the formula we obtain

$$\begin{aligned}
 &a_1u(x) + a_2u(x-h) + a_3u\left(x - \frac{1}{2}h\right) \\
 &= a_1u(x) + a_2\left(u(x) - hu'(x) + \frac{h^2}{2}u''(x) - \frac{h^3}{6}u'''(x) + O(h^4)\right) \\
 &\quad + a_3\left(u(x) - \frac{h}{2}u'(x) + \frac{h^2}{8}u''(x) - \frac{h^3}{8}u'''(x) + \frac{1}{6}\left(\frac{h}{2}\right)^3u'''(x) + O(h^4)\right) \\
 &= (a_1 + a_2 + a_3)u(x) + \left(-a_2h - a_3\frac{h}{2}\right)u'(x) + \left(a_2\frac{h^2}{2} + a_3\frac{h^2}{8}\right)u''(x) \\
 &\quad + \frac{h^3}{6}\left(-a_2 - a_3\left(\frac{1}{2}\right)\right)u'''(x) + O(h^3).
 \end{aligned}$$

We want this expression to be an approximation for the second derivative, thus we have to impose

$$\begin{cases} a_1 + a_2 + a_3 = 0 \\ -a_2h - a_3\frac{h}{2} = 0 \\ a_2\frac{h^2}{2} + a_3\frac{h^2}{8} = h^2 \end{cases}$$

The last requirement is imposed so that, once divided by h^2 , we recover the second derivative. Solving the system leads to

$$a_1 = 4, \quad a_2 = 4, \quad a_3 = -8$$

Our approximation thus reads

$$\frac{1}{h^2} \left(4u(x) + 4u(x-h) - 8u\left(x + \frac{1}{2}h\right) \right) = u''(x) - \overbrace{\frac{1}{2}hu'''(x)}^{e(h)} + O(h^2).$$

Problem 4. (A boundary value problem)

Given the two point boundary value problem:

$$u_{xx} + 2u_x + \pi^2 u = \cos(\pi x) - \pi(x+1)\sin(\pi x), \quad 0 \leq x \leq 2, \quad u(0) = 0, \quad u(2) = 1$$

- a) Verify that the exact solution is

$$u(x) = \frac{x}{2} \cos(\pi x).$$

- b) Set up a finite difference scheme for this problem, using central differences. Use $\Delta x = 2/N$ as the grid size, and let $x_i = i\Delta x$, $i = 0, 1, \dots, N$.
- c) Let $N = 4$ and use the above formula to find approximations $U_i \approx u(x_i)$, $i = 1, 2, 3$. (That is: Set up the system of equations, and solve it). Compare with the exact solution.
- d) (J) Modify the code Example 1, BVP in the note on boundary value problems, and solve the problem numerically. Use $N = 10, 20, 40$ in your simulation. For each N , write down the error

$$e(h) = \max_{i=0, \dots, N} |u(x_i) - U_i|.$$

What can you deduce about the order of the scheme from this experiment?

Solution.

- a) First we compute the single derivatives

$$u_x = \frac{\cos(\pi x)}{2} - \frac{\pi x}{2} \sin(\pi x)$$

$$u_{xx} = -\frac{\pi}{2} \sin(\pi x) - \frac{\pi}{2} \sin(\pi x) - \frac{\pi^2 x}{2} \cos(\pi x)$$

Plugging together we obtain

$$\begin{aligned}
 u_{xx} + 2u_x + \pi^2 u &= -\pi \sin(\pi x) - \cancel{\frac{\pi^2 x}{2} \cos(\pi x)} + \cos(\pi x) - \pi x \sin(\pi x) + \cancel{\frac{\pi^2 x}{2} \cos(\pi x)} \\
 &= \cos(\pi x) - \pi(x+1) \sin(\pi x)
 \end{aligned}$$

b) Using central differences and using $\Delta x = h$, I obtain

$$\begin{aligned}
 &\frac{U^{i+1} - 2U^i + U^{i-1}}{h^2} + 2\frac{U^{i+1} - U^{i-1}}{2h} + \pi^2 U^i \\
 &= \left[\frac{1}{h^2} - \frac{1}{h} \right] U^{i-1} + \left[\pi^2 - \frac{2}{h^2} \right] U^i + \left[\frac{1}{h^2} + \frac{1}{h} \right] U^{i+1} = f^i
 \end{aligned}$$

where

$$f^i = \cos(i\pi h) - \pi(ih+1) \sin(i\pi h)$$

Since we know from the Dirichlet boundary conditions that $U^0 = 0$ and $U^N = 1$ we have the modified expressions:

$$\begin{aligned}
 &\left[\frac{1}{h^2} - \frac{1}{h} \right] 0 + \left[\pi^2 - \frac{2}{h^2} \right] U^1 + \left[\frac{1}{h^2} + \frac{1}{h} \right] U^2 = f^1 \\
 &\left[\frac{1}{h^2} - \frac{1}{h} \right] U^{N-2} + \left[\pi^2 - \frac{2}{h^2} \right] U^{N-1} + \left[\frac{1}{h^2} + \frac{1}{h} \right] 1 = f^{N-1}
 \end{aligned}$$

that become

$$\begin{aligned}
 &\left[\pi^2 - \frac{2}{h^2} \right] U^1 + \left[\frac{1}{h^2} + \frac{1}{h} \right] U^2 = f^1 \\
 &\left[\frac{1}{h^2} - \frac{1}{h} \right] U^{N-2} + \left[\pi^2 - \frac{2}{h^2} \right] U^{N-1} = f^{N-1} - \left[\frac{1}{h^2} + \frac{1}{h} \right]
 \end{aligned}$$

c) The resulting system is of the form $Ax = b$ where A is a tridiagonal matrix in the form

$$A = \begin{bmatrix} \pi^2 - \frac{2}{h^2} & \frac{1}{h^2} + \frac{1}{h} & 0 & 0 \\ \frac{1}{h^2} - \frac{1}{h} & \ddots & \frac{1}{h^2} + \frac{1}{h} & 0 \\ 0 & \ddots & \ddots & \ddots \end{bmatrix}$$

and

$$x = \begin{bmatrix} U^1 \\ \vdots \\ U^{N-1} \end{bmatrix}, \quad b = \begin{bmatrix} f^1 \\ f^2 \\ \vdots \\ f^{N-2} \\ f^{N-1} - \left[\frac{1}{h^2} + \frac{1}{h} \right] \end{bmatrix}$$

d) see the Jupyter notebook `bvp-linear-d-complete.ipynb`

Optional Exercises.

The next two exercises are optional and should not be handed in

Problem 5. (A boundary value problem with variable coefficients)

Given the two point boundary value problem:

$$u_{xx} - \frac{2}{x}u_x + \frac{2}{x^2}u = -\frac{x\pi^2}{2}\cos(\pi x), \quad 1 \leq x \leq 2, \quad u(1) = -\frac{1}{2} \quad u(2) = 1$$

a) Verify that the exact solution is

$$u(x) = \frac{x}{2}\cos(\pi x).$$

b) Set up a finite difference scheme for this problem, using central differences. Use $\Delta x = 1/N$ as the grid size, and let $x_i = i\Delta x$, $i = 0, 1, \dots, N$.

c) Let $N = 4$ and use the above formula to find approximations $U_i \approx u(x_i)$, $i = 1, 2, 3$. (That is: Set up the system of equations, and solve it). Compare with the exact solution.

d) (J) Modify the code Example 1, BVP in the note on boundary value problems, and solve the problem numerically. Use $N = 10, 20, 40$ in your simulation. For each N , write down the error

$$e(h) = \max_{i=0, \dots, N} |u(x_i) - U_i|.$$

What can you deduce about the order of the scheme from this experiment?

Solution.

a) First we compute the single derivatives

$$u_x = \frac{\cos(\pi x)}{2} - \frac{\pi x}{2}\sin(\pi x)$$

$$u_{xx} = -\frac{\pi}{2}\sin(\pi x) - \frac{\pi}{2}\sin(\pi x) - \frac{\pi^2 x}{2}\cos(\pi x)$$

Plugging together we obtain

$$\begin{aligned}
 u_{xx} - \frac{2}{x}u_x + \frac{2}{x^2}u &= \cancel{-\pi \sin(\pi x)} - \frac{\pi^2 x}{2} \cos(\pi x) - \frac{1}{x} \cancel{\cos(\pi x)} + \cancel{\pi \sin(\pi x)} + \frac{1}{x} \cancel{\cos(\pi x)} \\
 &= -\frac{\pi^2 x}{2} \cos(\pi x)
 \end{aligned}$$

b) Using central differences and using $\Delta x = h$, I obtain

$$\begin{aligned}
 &\frac{U^{i+1} - 2U^i + U^{i-1}}{h^2} - \frac{2}{1+ih} \frac{U^{i+1} - U^{i-1}}{2h} + \frac{2}{(1+ih)^2} U^i \\
 &= \left[\frac{1}{h^2} + \frac{1}{h+ih^2} \right] U^{i-1} + \left[\frac{2}{(1+ih)^2} - \frac{2}{h^2} \right] U^i + \left[\frac{1}{h^2} - \frac{1}{h+ih^2} \right] U^{i+1} = f^i
 \end{aligned}$$

where

$$f^i = -\frac{i\pi^2 h}{2} \cos(i\pi h)$$

Since we know from the Dirichlet boundary conditions that $U^0 = -1/2$ and $U^N = 1$ we have the modified expressions:

$$\left[\frac{1}{h^2} + \frac{1}{h+ih^2} \right] \cdot \left(-\frac{1}{2} \right) + \left[\frac{2}{(1+h)^2} - \frac{2}{h^2} \right] U^1 + \left[\frac{1}{h^2} - \frac{1}{h+h^2} \right] U^2 = f^1$$

and

$$\begin{aligned}
 &\left[\frac{1}{h^2} + \frac{1}{h+(N-1)h^2} \right] U^{N-2} + \left[\frac{2}{(1+(N-1)h)^2} - \frac{2}{h^2} \right] U^{N-1} \\
 &\quad + \left[\frac{1}{h^2} - \frac{1}{h+(N-1)h^2} \right] 1 = f^{N-1}
 \end{aligned}$$

that become

$$\left[\frac{2}{(1+h)^2} - \frac{2}{h^2} \right] U^1 + \left[\frac{1}{h^2} - \frac{1}{h+h^2} \right] U^2 = f^1 + \frac{1}{2} \left[\frac{1}{h^2} + \frac{1}{h+h^2} \right]$$

and

$$\begin{aligned}
 &\left[\frac{1}{h^2} + \frac{1}{h+(N-1)h^2} \right] U^{N-2} + \left[\frac{2}{(1+(N-1)h)^2} - \frac{2}{h^2} \right] U^{N-1} \\
 &\quad = f^{N-1} - \frac{1}{h^2} + \frac{1}{h+(N-1)h^2}
 \end{aligned}$$

c) The resulting system is of the form $Ax = b$ where A is a tridiagonal matrix in the form

$$A = \begin{bmatrix} \left[\frac{2}{(1+h)^2} - \frac{2}{h^2} \right] & \left[\frac{1}{h^2} - \frac{1}{h+h^2} \right] & 0 & 0 \\ \left[\frac{1}{h^2} + \frac{1}{h+2h^2} \right] & \left[\frac{2}{(1+2h)^2} - \frac{2}{h^2} \right] & \left[\frac{1}{h^2} - \frac{1}{h+2h^2} \right] & 0 \\ 0 & \ddots & \ddots & \ddots \end{bmatrix}$$

and

$$x = \begin{bmatrix} U^1 \\ \vdots \\ U^{N-1} \end{bmatrix}, \quad b = \begin{bmatrix} f^1 + \frac{1}{2} \left[\frac{1}{h^2} + \frac{1}{h+h^2} \right] \\ f^2 \\ \vdots \\ f^{N-2} \\ f^{N-1} - \frac{1}{h^2} + \frac{1}{h+(N-1)h^2} \end{bmatrix}$$

d) see Ex1Pr5.ipynb

Problem 6. (a *nonlinear* boundary value problem)

Given the two point boundary value problem:

$$\frac{1}{10}u'' + u' + u^2 = 0, \quad u(0) = 1, \quad u(1) = 0.$$

- Set up a finite difference scheme for this problem, using central differences. Use $\Delta x = 1/N$ as the grid size, and let $x_i = i\Delta x$, $i = 0, 1, \dots, N$. You end up with a system of nonlinear equations.
- (J). Solve the problem in python. Use $N = 50$ (although you may try other values as well).

Hint: Use the python function `scipy.optimize.fsolve`. Read the documentation to figure out how to use it.

Solution.

a) The set of nonlinear equations to solve is simply

$$0.1 \frac{U^{i+1} - 2U^i + U^{i-1}}{h^2} + \frac{U^{i+1} - U^{i-1}}{2h} + (U^i)^2, \quad i = 1, \dots, N-1$$

with $U_0 = 1$ and $U_N = 0$.

b) see Ex1Pr6.ipynb